

Berenstein–Zelevinsky Polytopes for Type- G_2 Lie Algebras

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Introduction

Let $\mathfrak{g}$ be a finite-dimensional complex semisimple Lie algebra of rank r .

In [1], Berenstein and Zelevinsky defined a polytope $BZ_{\lambda\nu}^{\mu}(\mathfrak{g})$ in which the number of integer lattice points equals the multiplicity with which the irreducible representation V_{μ} appears in the tensor product $V_{\lambda} \otimes V_{\nu}$. The definition of $BZ_{\lambda\nu}^{\mu}(\mathfrak{g})$ is given in terms of so-called **i-trails**.

For the classical types A, B, C, D , Berenstein and Zelevinsky computed these **i-trails** and gave *completely explicit* linear inequalities for $BZ_{\lambda\nu}^{\mu}(\mathfrak{g})$. However, this remains to be done for the exceptional types G_2, F_4, E_6, E_7, E_8 .

Our Contribution

Let $\mathfrak{g}$ be of type G_2 .

- We compute the **i-trails** required to define $BZ_{\lambda\nu}^{\mu}(\mathfrak{g})$.
- We give *completely explicit inequalities* for $BZ_{\lambda\nu}^{\mu}(\mathfrak{g})$, implemented in **SageMath**.
- We study the *geometry* and *Ehrhart theory* of $BZ_{\lambda\nu}^{\mu}(\mathfrak{g})$.

Definition of i-trails

Let γ and δ be weights of an irreducible representation V_{λ} of $\mathfrak{g}$, and let $\mathbf{i} = (i_1, \dots, i_{\ell})$ be a sequence of indices in $[r]$. An **i-trail** from γ to δ is a sequence $\gamma = \gamma_0, \gamma_1, \dots, \gamma_{\ell} = \delta$ of weights of V_{λ} such that:

- **The Combinatorial Rule:** for $k \in [\ell]$, $\gamma_{k-1} - \gamma_k = c_k \alpha_{i_k}$ for some nonnegative c_k .
- **The Composition Rule:** $e_{i_1}^{c_1} \cdots e_{i_{\ell}}^{c_{\ell}}$ is a non-zero map $V(\delta) \rightarrow V(\gamma)$, where the e_{i_k} are Chevalley generators.

A *potential i-trail* is a sequence of weights of V_{λ} satisfying the combinatorial rule but not necessarily the composition rule.

Examples

Let $\mathfrak{g}$ be of type G_2 . Then $BZ_{\lambda\nu}^{\mu}(\mathfrak{g})$ is a polytope of dimension ≤ 4 in $\mathbb{R}^6$.

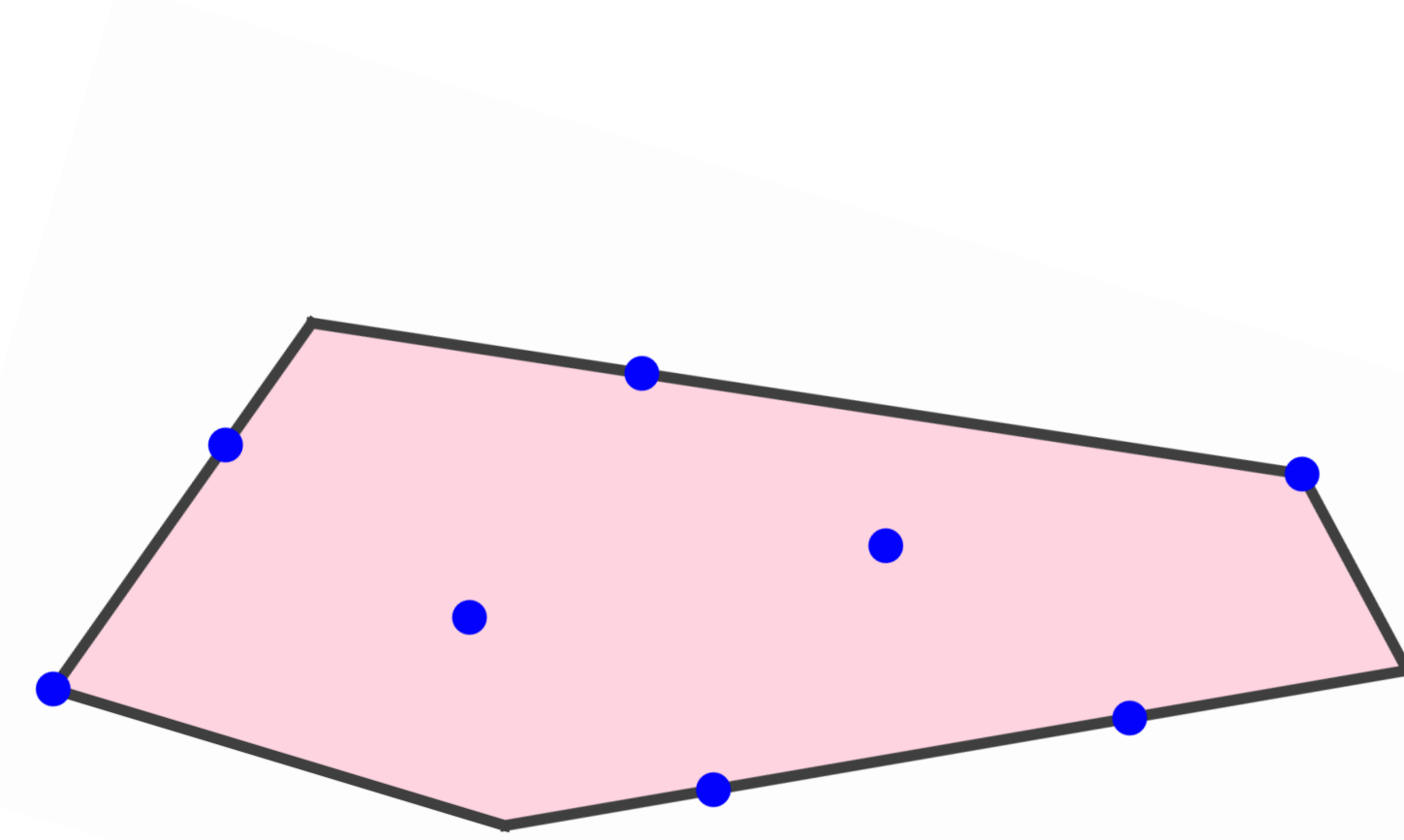


Figure 1: Let $\lambda = (10, 0)$, $\nu = (8, 3)$, and $\mu = (0, 8)$. Then $BZ_{\lambda\nu}^{\mu}(\mathfrak{g})$ is a 2-dimensional polytope with 8 lattice points, so $c_{\lambda,\nu}^{\mu} = 8$.

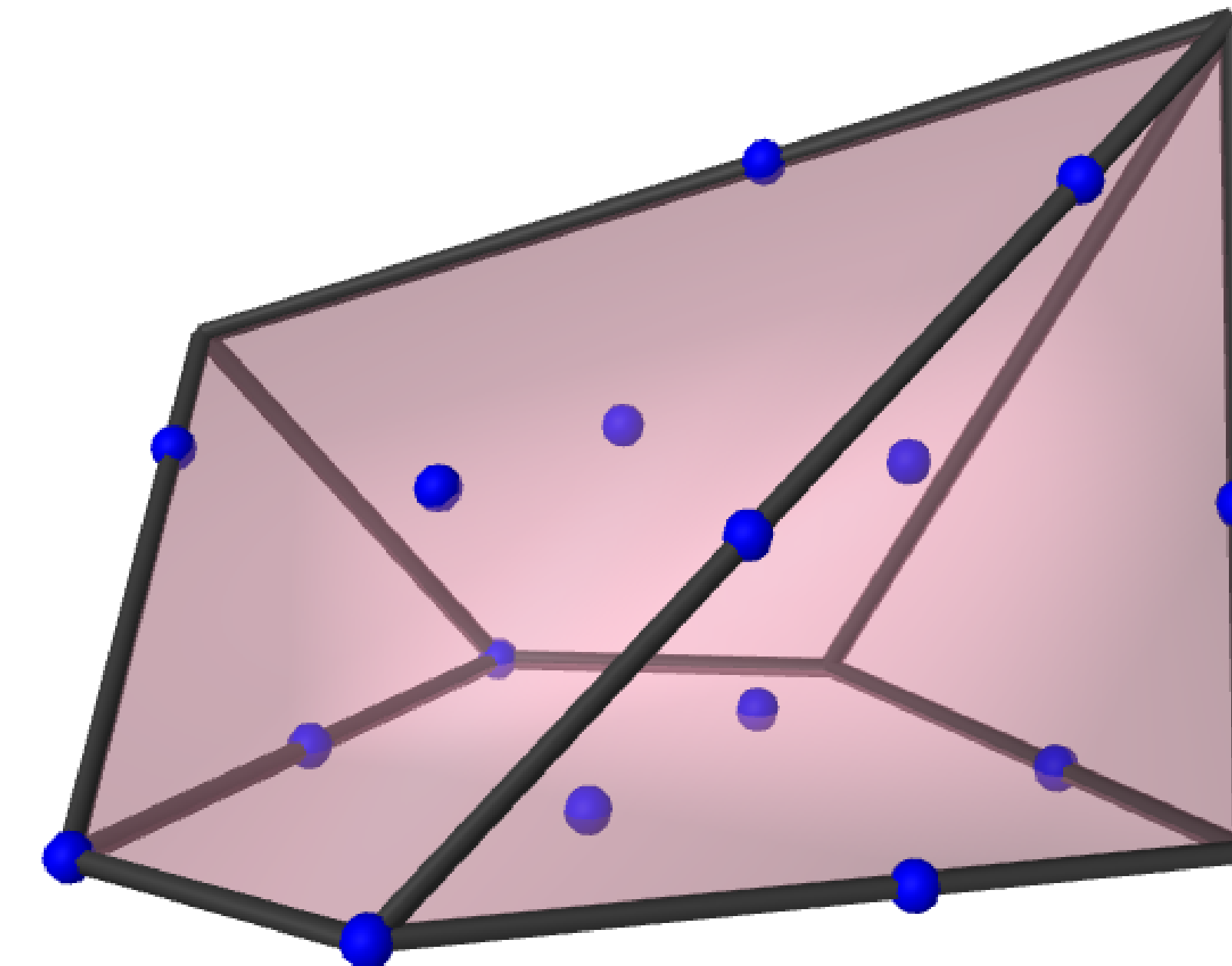
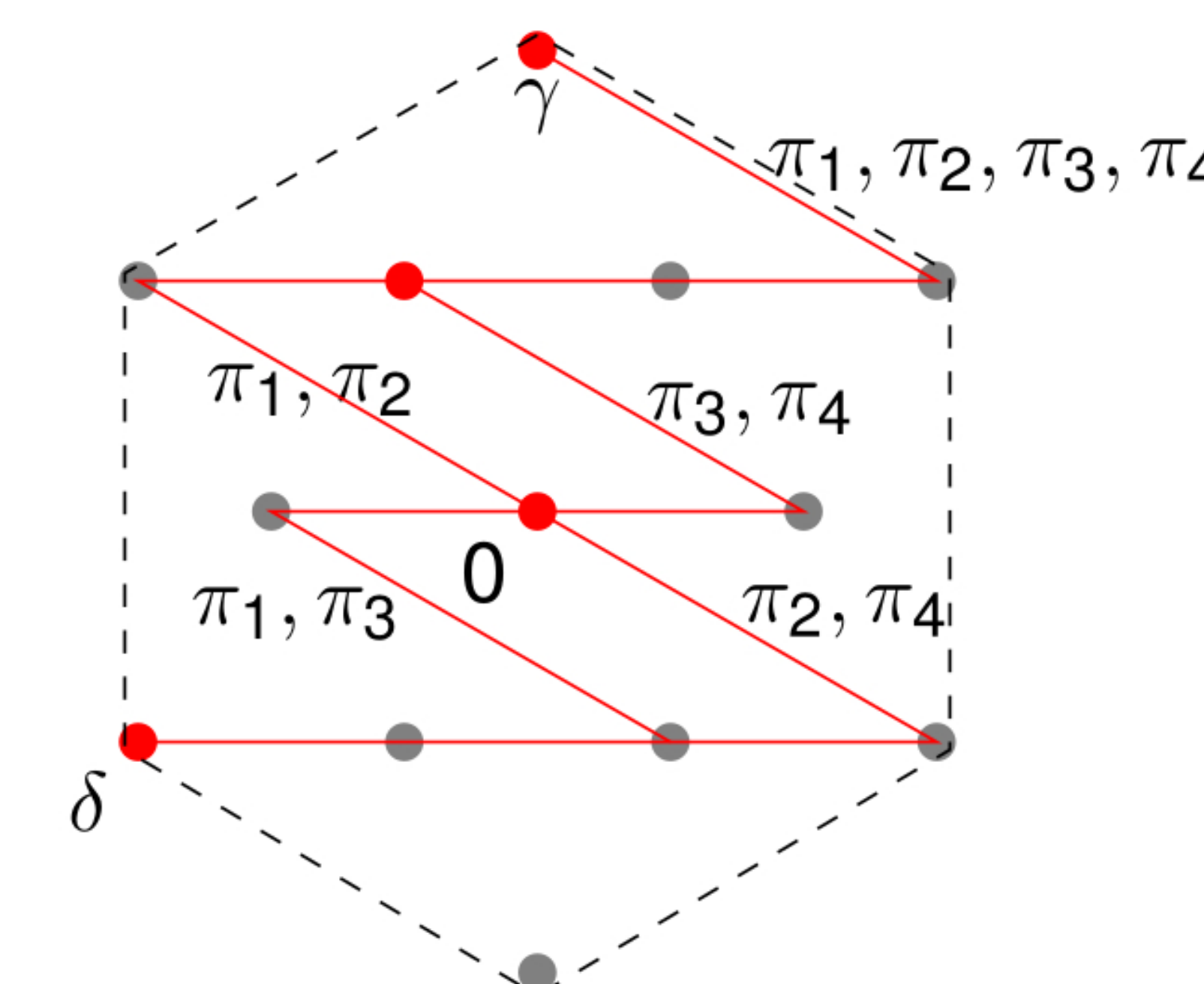
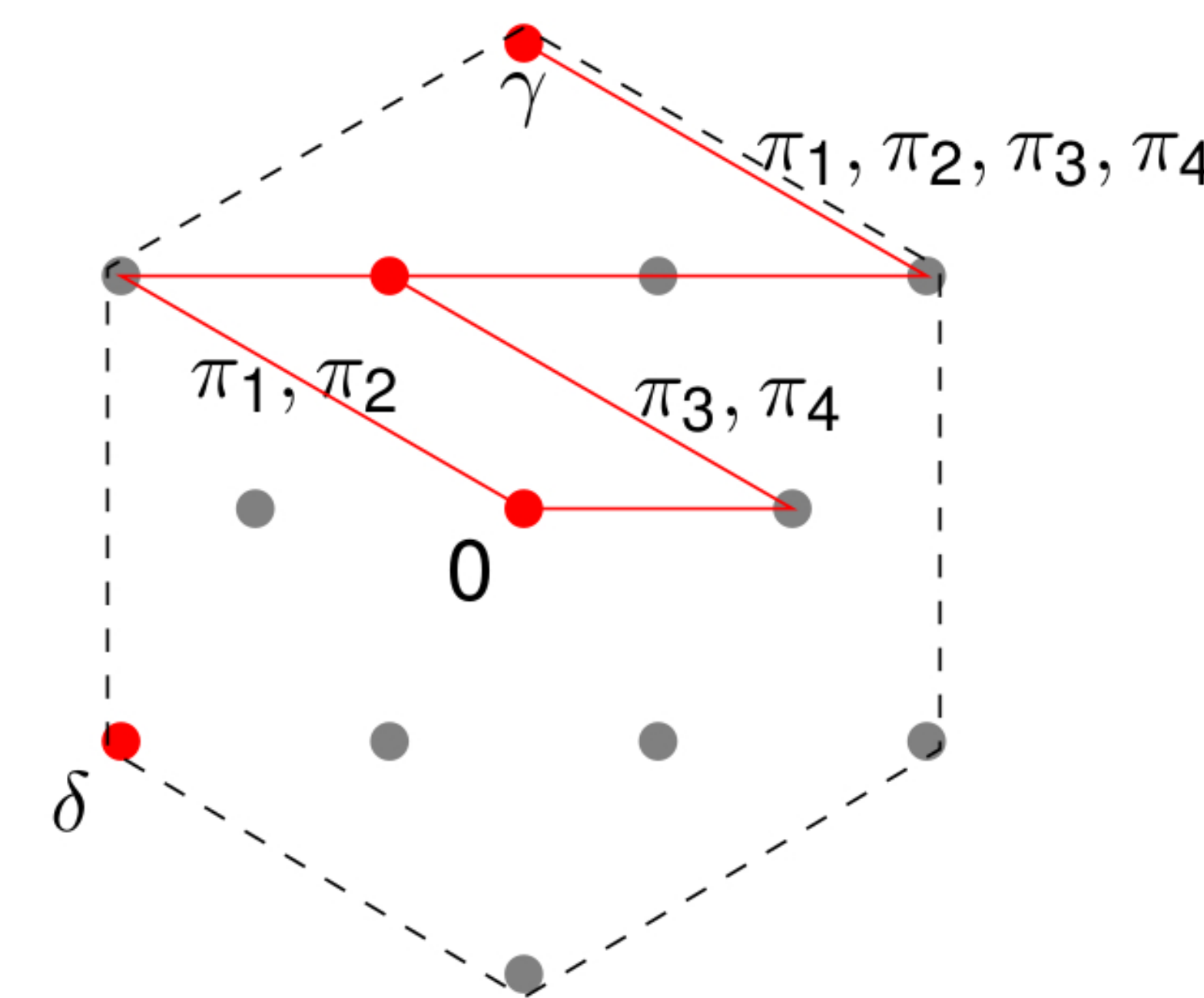
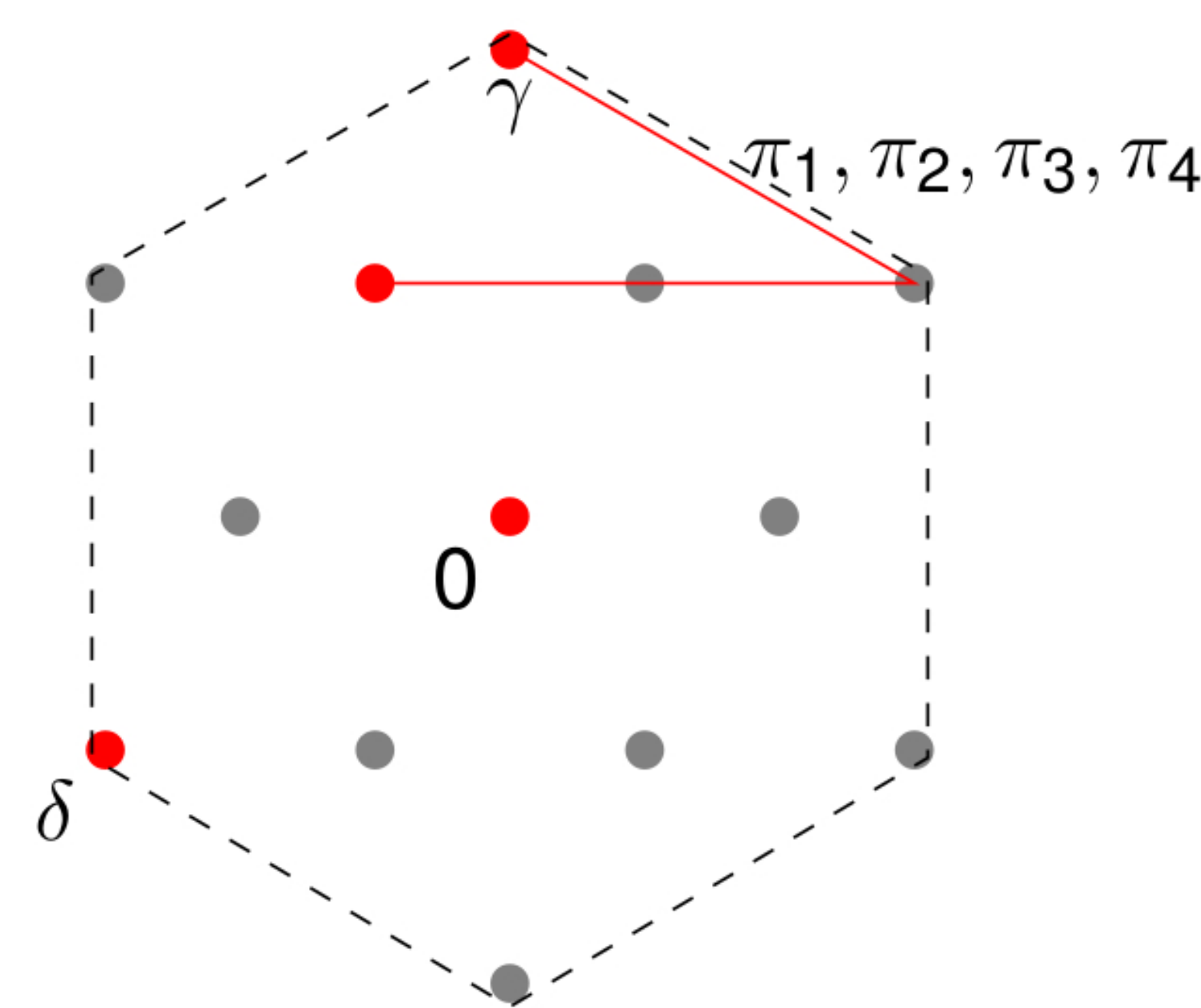


Figure 2: Let $\lambda = (6, 0)$, $\mu = (8, 6)$, and $\nu = (9, 6)$. Then $BZ_{\lambda\nu}^{\mu}(\mathfrak{g})$ is a 3-dimensional polytope with 16 lattice points, so $c_{\lambda,\nu}^{\mu} = 16$.

Computing the i-trails

The *combinatorial rule* defines a polytope in which the lattice points correspond to *potential i-trail*.

Let $\mathbf{i} = (1, 2, 1, 2, 1, 2)$. Here is a step-by-step construction of the *underlying paths* $\pi_1, \pi_2, \pi_3, \pi_4$ that support the **i-trails** between the indicated weights γ and δ in the irreducible representation V_{γ} for type G_2 .



Results

Theorem

Let λ , μ , and ν be dominant weights of a Lie algebra $\mathfrak{g}$ of type G_2 .

- All potential **i-trails** involved in the definition of $BZ_{\lambda\nu}^{\mu}(\mathfrak{g})$ satisfy the composition rule.
- The possible vertex denominators in $BZ_{\lambda\nu}^{\mu}(\mathfrak{g})$ are 1, 2, 3, 4, 5, 6, and 9. The largest possible denominator of $BZ_{\lambda\nu}^{\mu}(\mathfrak{g})$ is 180.
- The Ehrhart quasi-period of $BZ_{\lambda\nu}^{\mu}(\mathfrak{g})$ divides 6. In fact, each of the divisors 1, 2, 3, 6 appears as a quasi-period of a type- G_2 BZ polytope.

The Ehrhart-positivity conjecture fails in type G_2

In types A, B, C, D , the Ehrhart quasipolynomial of $BZ_{\lambda\nu}^{\mu}(\mathfrak{g})$ is conjectured to be Ehrhart positive [2], meaning that all coefficients of their Ehrhart quasipolynomials are nonnegative

However, many type- G_2 BZ polytopes have Ehrhart quasipolynomials with negative coefficients. Examples appear in every possible dimension. Empirically, it seems that roughly 10% of the type- G_2 BZ polytopes with the coordinates of the dominant weights less than 10,000 have negative coefficients.

References

- [1] Arkady D. Berenstein and Andrei V. Zelevinsky. Tensor product multiplicities, canonical bases and totally positive varieties. *Invent. Math.*, 143(1):77–128, 2001.
- [2] Jesús A. De Loera and Tyrrell B. McAllister. On the computation of Clebsch-Gordan coefficients and the dilation effect. *Experiment. Math.*, 15(1):7–19, 2006.